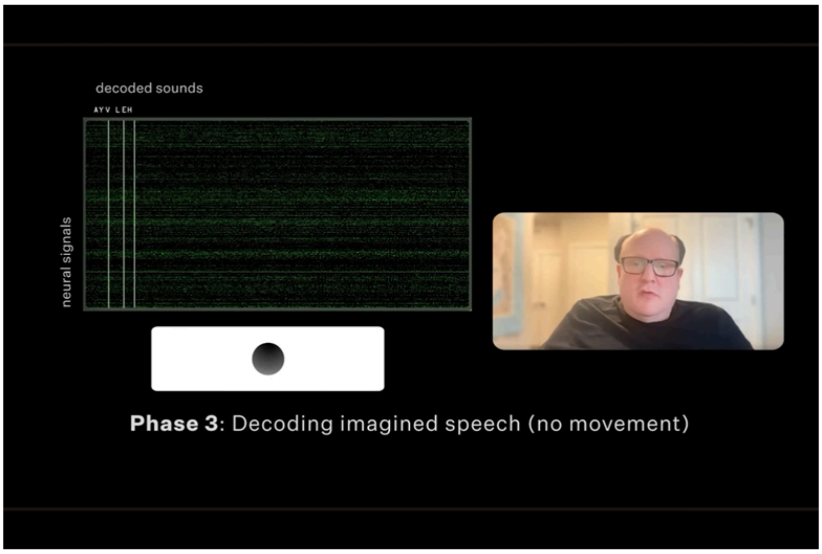
Restoring Speech to Paralyzed Patients: The History of Speech BCI

Can paralyzed patients speak without opening their mouths, just by thinking? This deck walks through the history of invasive speech brain-computer interfaces, from the discovery of high-gamma signals to the current stage of long-term implantation and imagined speech.

Neuralink – Imagined Speech

Bruce Yixuan Li



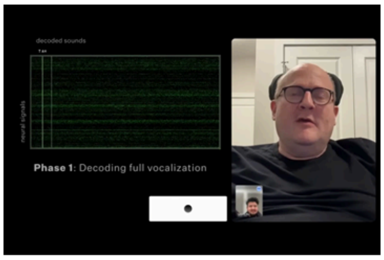
https://www.youtube.com/@neuralink

2026/4/6	2026	1999-2001	2009-2020	2021-2026	202?-	Page: 1

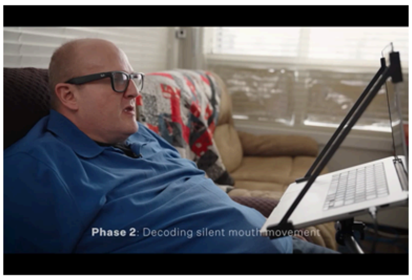
Hello, I'm Yixuan. Can paralyzed patients speak without opening their mouths, just by thinking? Are invasive speech BCIs really that powerful? On March 25, Neuralink released a new video showing exactly this process.

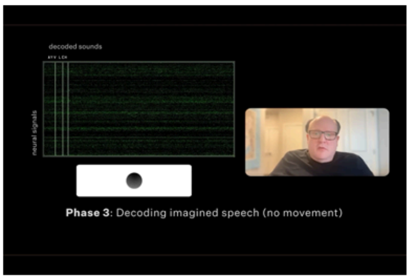
Read Aloud -> Mouth -> Imagined

Bruce Yixuan Li



- **Which Signal?**
- **Which Model?**
- **How To Prove Safety?**



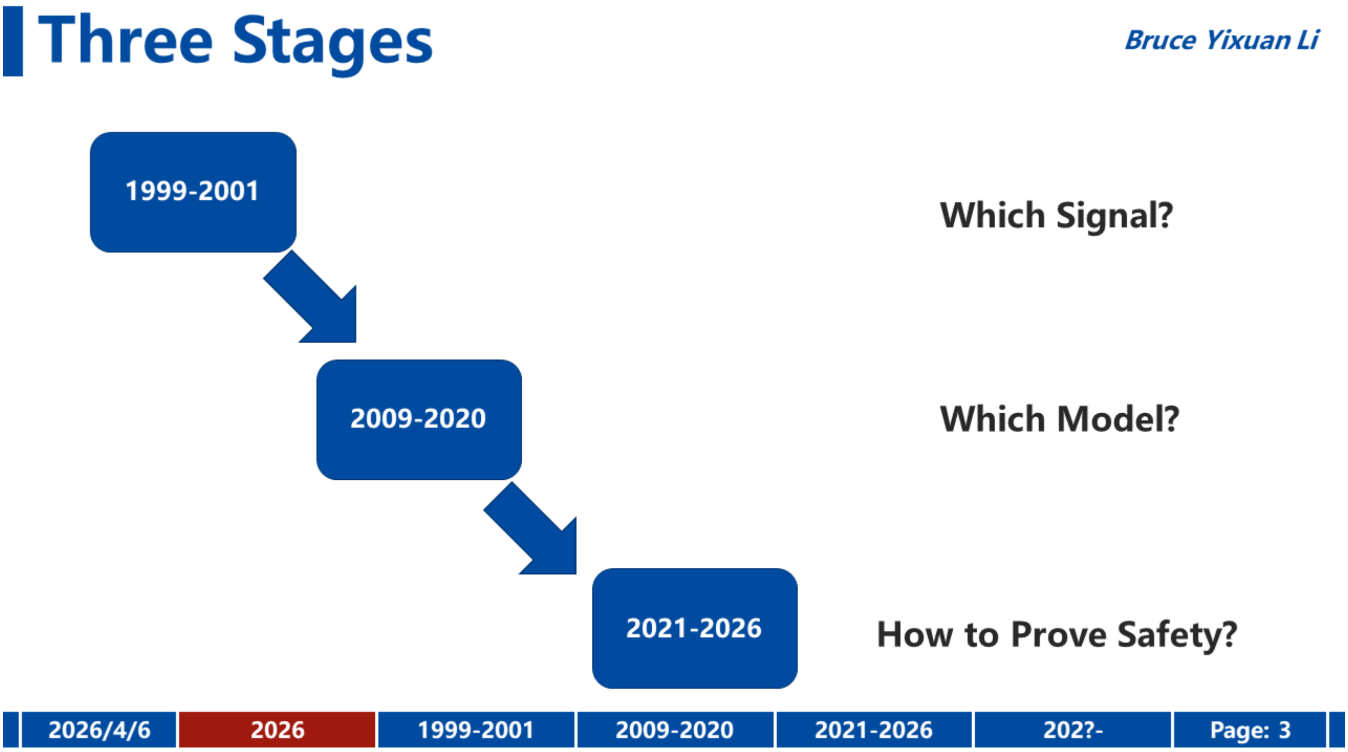


https://www.youtube.com/@neuralink

2026/4/6	2026	1999-2001	2009-2020	2021-2026	202?-	Page: 2

Of course, this result was not achieved in a single step. The training had three stages: first, the patient read the text on the screen aloud. Second, the patient silently mouthed the text on the screen. Third, the patient simply "thought" the words in their mind. That is what we just saw.

The model for each stage had to be trained before moving to the next one. If you think carefully about this process, there are three problems to solve here. First, what signal should be decoded? Second, what model should be used for decoding? Third, how do we prove the safety of the BCI?



In fact, the development of speech BCIs has unfolded in exactly these three stages: first, figuring out what signal to decode; then, figuring out what model to use; and now, reaching the stage of verifying long-term implantation safety.

Next, I'll walk you through the history of speech BCIs. If you only care about recent progress, you can jump straight to Chapter 4.

Stage 1: 1999-2001, which signal to decode?

Which Signal?

Bruce Yixuan Li

Brain (1998), **121,** 2301–2315

Functional mapping of human sensorimotor cortex with electrocorticographic spectral analysis
II. Event-related synchronization in the gamma band

Nathan E. Crone,[1] Diana L. Miglioretti,[3] Barry Gordon[1,4,5] and Ronald P. Lesser[1,2,5]

Departments of [1]Neurology and [2]Neurosurgery, The Johns Hopkins University School of Medicine, and the Departments of [3]Biostatistics and [4]Cognitive Science, and [5]Zanvyl Krieger Mind/Brain Institute, The Johns Hopkins University, Baltimore, Maryland, USA

Correspondence to: Dr Nathan Crone, Department of Neurology, The Johns Hopkins Hospital, Meyer 2–147, 600 N. Wolfe St, Baltimore, MD 21287, USA. E-mail: ncrone@jhmi.edu

Brazier Award-winning article, 2001

Induced electrocorticographic gamma activity during auditory perception

Nathan E. Crone[a,*], Dana Boatman[a,b], Barry Gordon[a,c], Lei Hao[a]

[a]*Department of Neurology, The Johns Hopkins University School Of Medicine, N. Wolfe St., Meyer Building, Baltimore, MD 21287-7247, USA*
[b]*Department of Otolaryngology, The Johns Hopkins University School Of Medicine, Baltimore, MD, USA*
[c]*The Zanvyl Krieger Mind/Brain Institute, The Johns Hopkins University, Baltimore, MD, USA*

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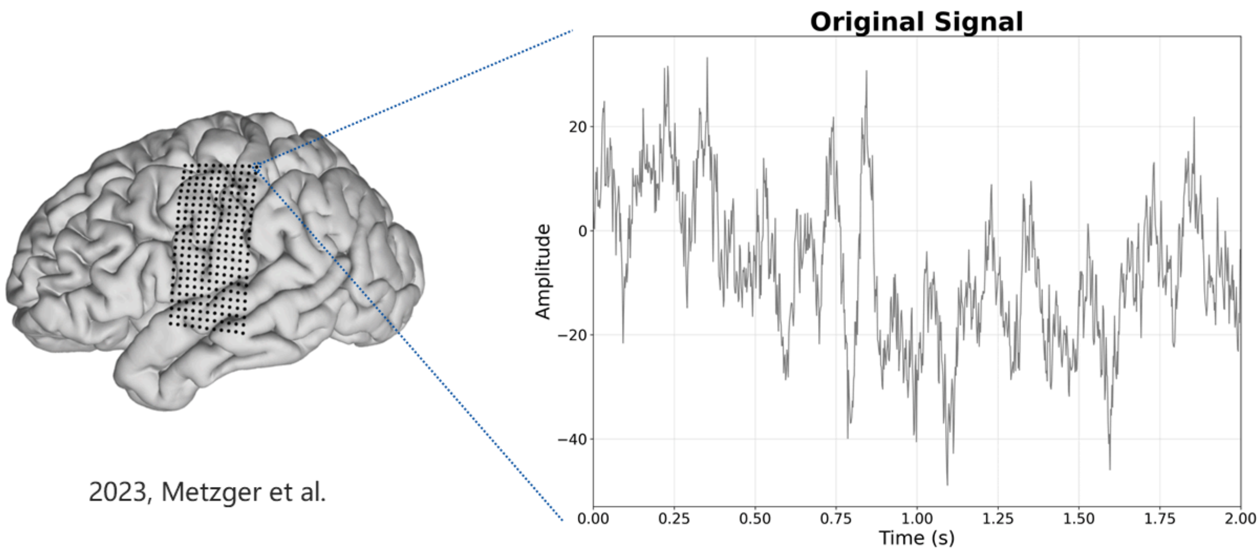
Let's start with two papers from 1998 and 2001. At that time, a U.S. team unexpectedly discovered that high-gamma signals in the human brain were directly related to heard vowels and consonants. One of the researchers, Nathan Crone, later became a key figure in this field.

Honestly, this was absolutely one of the most important discoveries in the entire history of brain-computer interfaces.

Meaning of High Gamma ?

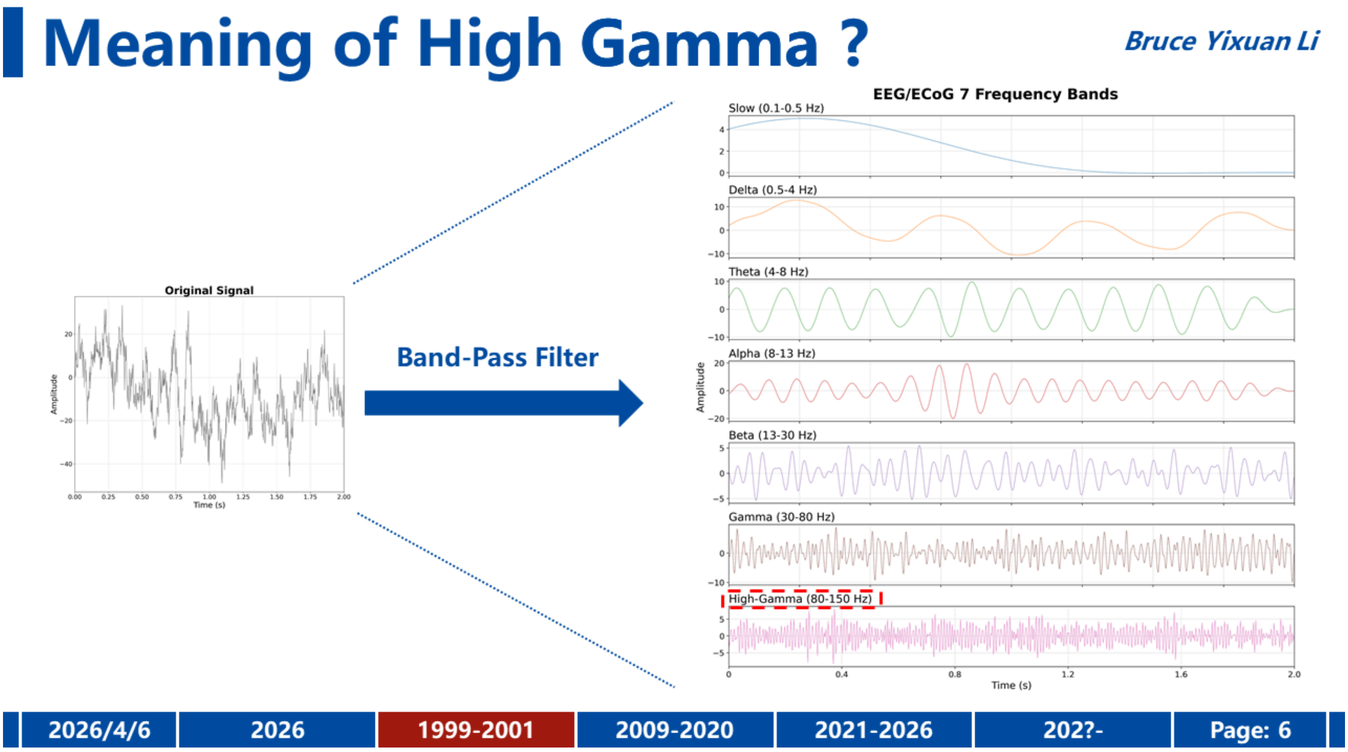
Bruce Yixuan Li



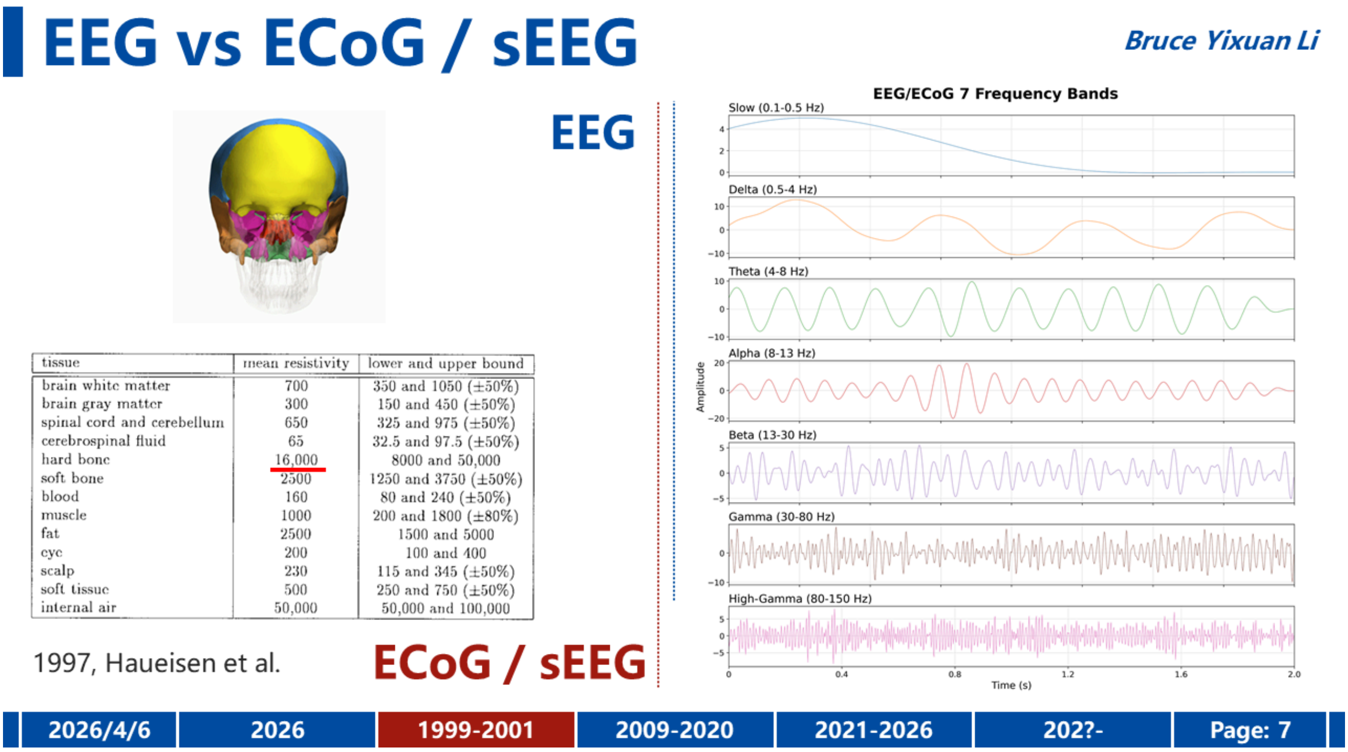
2023, Metzger et al.

2026/4/6	2026	1999-2001	2009-2020	2021-2026	202?-	Page: 5

What does the high-gamma signal mean? The signal recorded at each electrode site is itself a mixture of periodic and aperiodic components.



With a band-pass filter, we can separate it into different frequency bands. Researchers in the field conventionally divided it into seven bands. Any band above 80 Hz is what people in the field call the high-gamma signal.



tissue	mean resistivity	lower and upper bound
brain white matter	700	350 and 1050 (±50%)
brain gray matter	300	150 and 450 (±50%)
spinal cord and cerebellum	650	325 and 975 (±50%)
cerebrospinal fluid	65	32.5 and 97.5 (±50%)
hard bone	16,000	8000 and 50,000
soft bone	2500	1250 and 3750 (±50%)
blood	160	80 and 240 (±50%)
muscle	1000	200 and 1800 (±80%)
fat	2500	1500 and 5000
eye	200	100 and 400
scalp	230	115 and 345 (±50%)
soft tissue	500	250 and 750 (±50%)
internal air	50,000	50,000 and 100,000

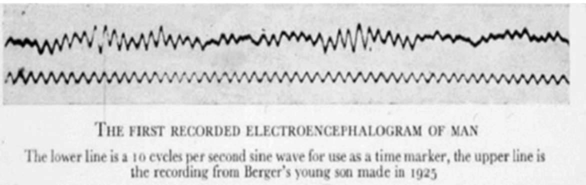
Scalp EEG, also called non-invasive EEG, can cover only 0.1 Hz to 80 Hz. Only intracranial EEG, also called invasive EEG, can capture signals above 80 Hz.

This is because the human skull has very high electrical resistance, nearly the highest of any tissue in the body, so only low-frequency signals can pass through it.

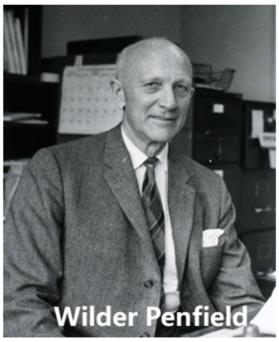
Why Delay

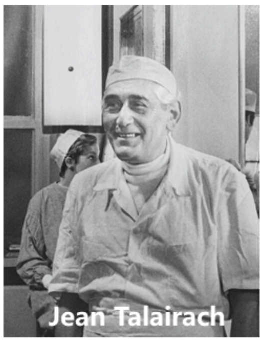
Bruce Yixuan Li

1924, EEG



1950s, ECoG & sEEG





1973, BCI

TOWARD DIRECT BRAIN-COMPUTER COMMUNICATION

JACQUES J. VIDAL[1]

Brain Research Institute,
University of California, Los Angeles, California

Why Was the Relationship Between High Gamma and Language Discovered So Late? (1998 or 2001)

That is why the relationship between high-gamma signals and language was discovered so late. Non-invasive EEG was invented in 1924; invasive EEG was developed in the 1950s by two surgeons; brain-computer interfaces were proposed in 1973. Yet the cornerstone of speech BCIs was not discovered until 1998.

Three Stages

Bruce Yixuan Li

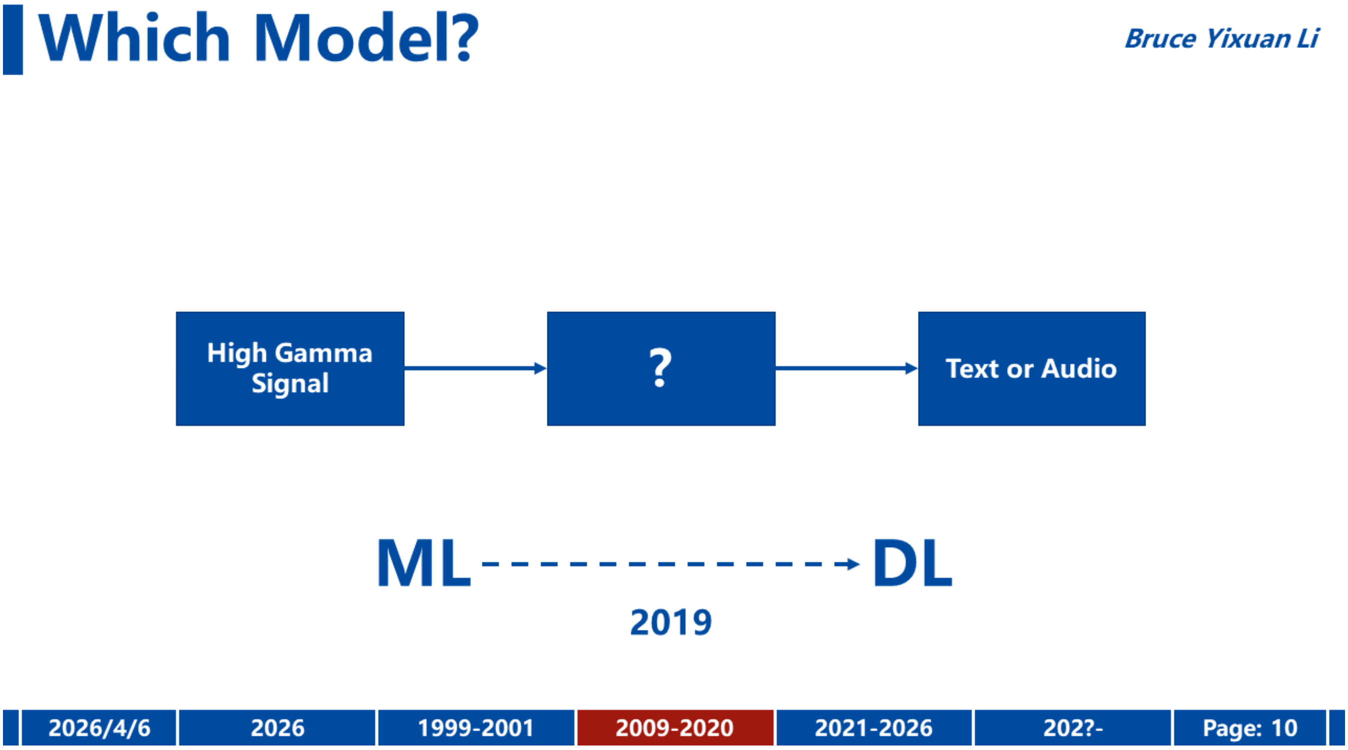
1999-2001 — Which Signal? --- High Gamma

2009-2020 — Which Model?

2021-2026 — How to Prove Safety?

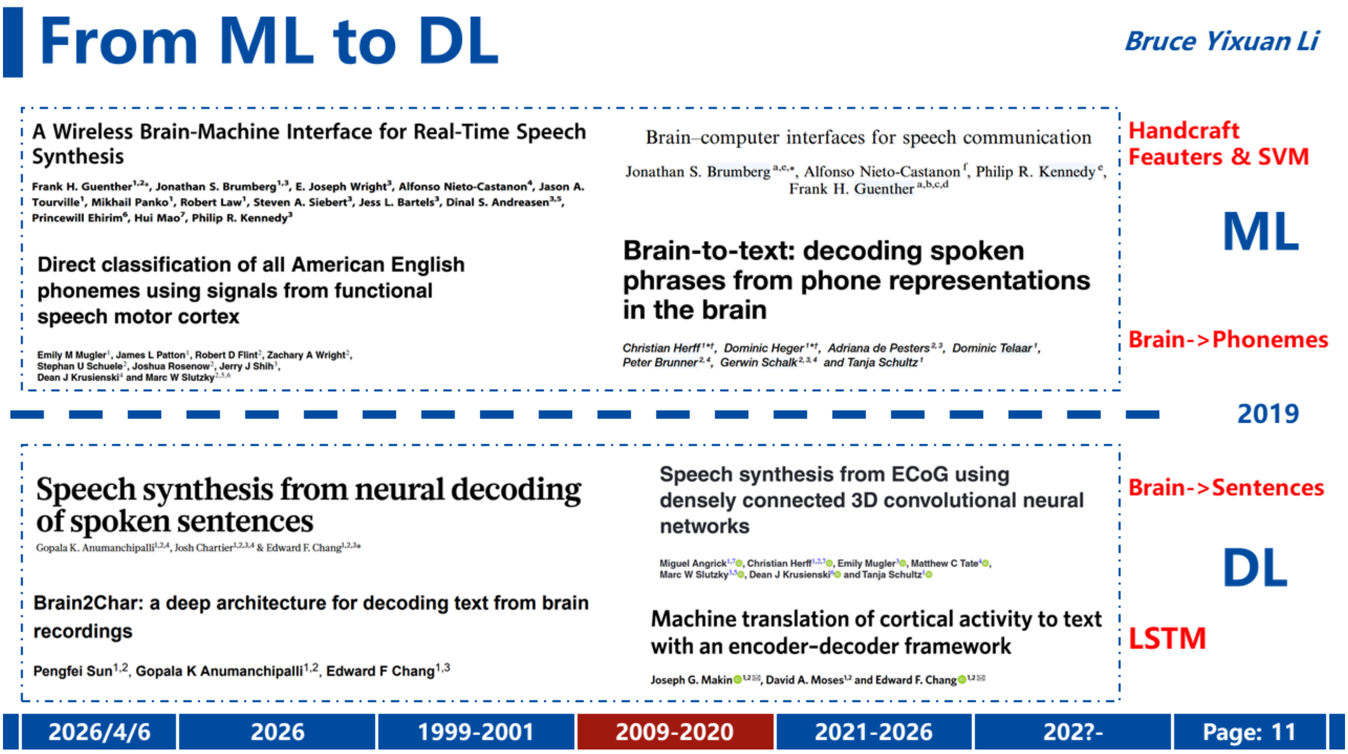
Now that humans know what signal to decode, the next step is to study what model to use for decoding.

Stage 2: 2009-2020, which model for decoding?



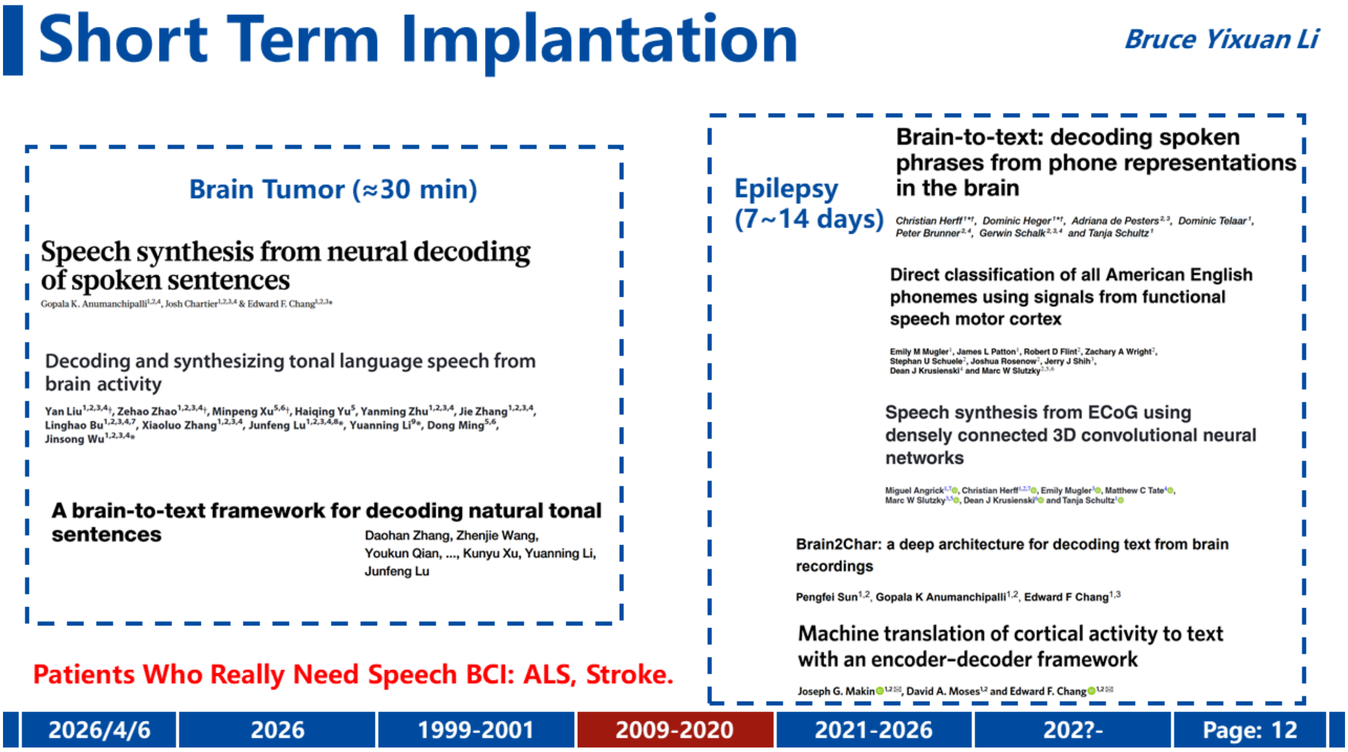
Now the input is a high-gamma time series, and the output is text or speech. So how do we complete this process?

Because the input signal is produced by a complex system composed of many units, and machine learning and deep learning are especially good at handling such complex systems. Like many other fields, our field also experienced a shift from machine learning to deep learning. In our field, that shift happened around 2019.

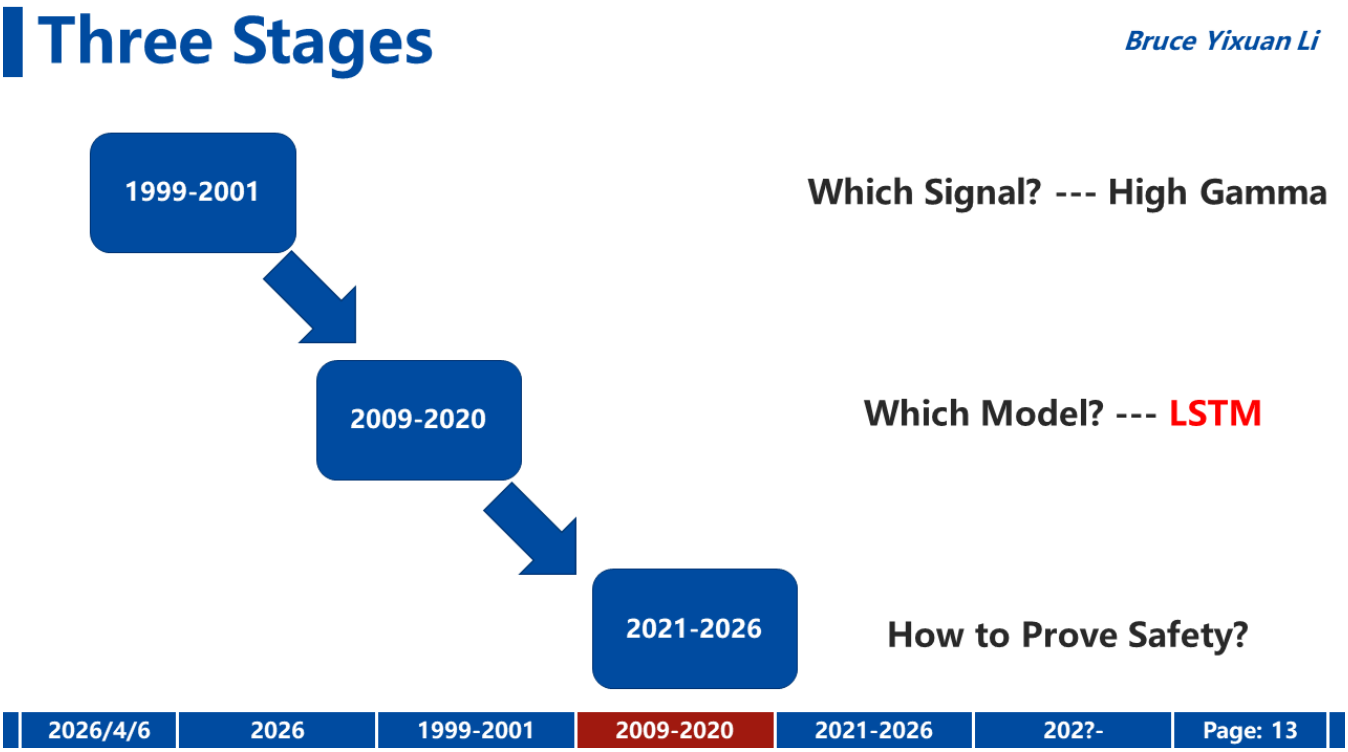


Before 2019, papers all relied on manually extracting features and then adding an SVM classifier. They could only decode phonemes. After 2019, sentence decoding became possible. The most effective model was the LSTM.

Many people ask: why didn't anyone try deep learning before 2019? You have to remember that AlexNet in 2012 was a hand-built CNN, and people even had to manually coordinate communication between two GPUs. Anyone without a computer science background simply couldn't manage that. It was not until TensorFlow in 2015 and PyTorch in 2016 that people no longer had to hand-build models. With about ten lines of code, you could set up an RNN, and the barrier to entry dropped dramatically.

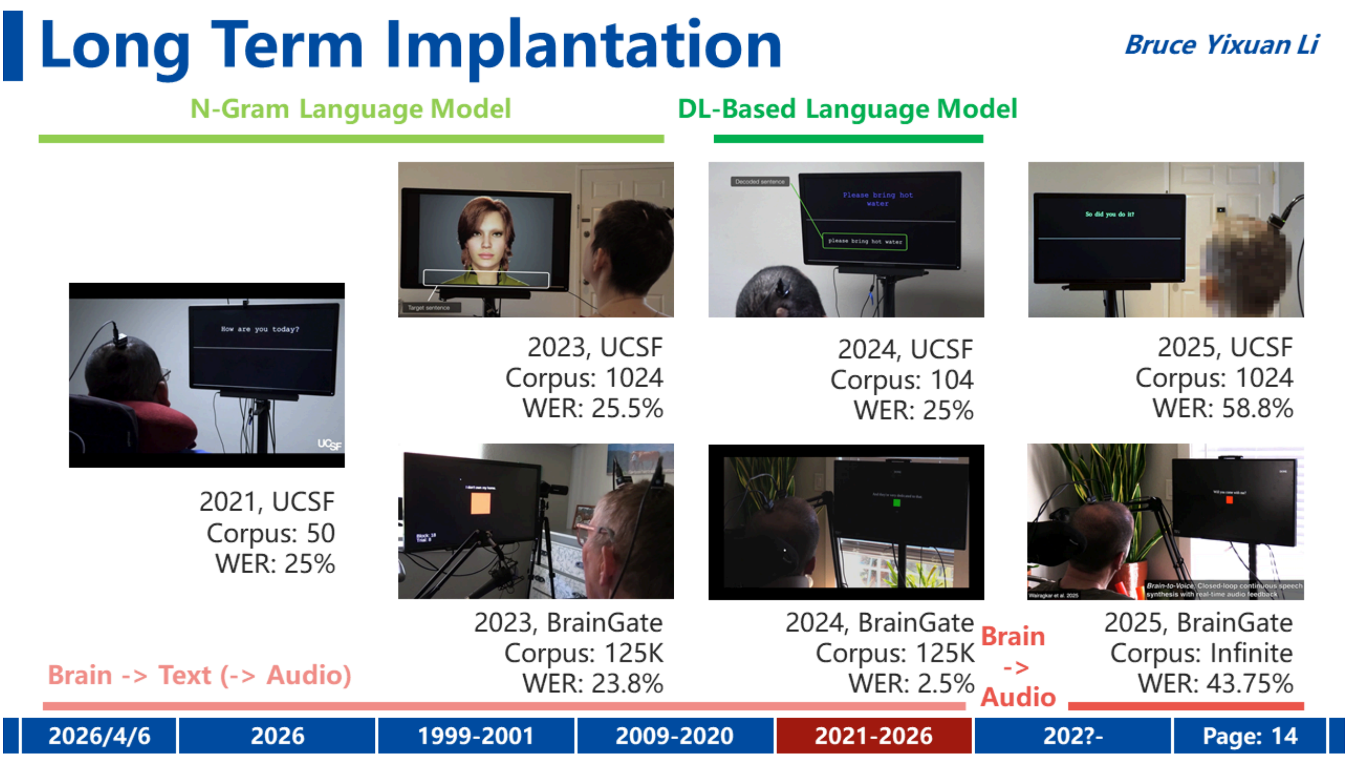


It should be pointed out that all of these studies were done in patients with short-term implants, usually people with brain tumors or epilepsy. But the patients who truly need speech BCIs are those with ALS or stroke.



Studies with short-term implants can tell us that LSTMs are effective, but they cannot tell us whether long-term implanted speech BCIs are safe.

Stage 3: 2021-2026, Safety of long term implantation



Finally, in 2021, the University of California, San Francisco (UCSF) produced the world's first long-term implanted speech BCI that could decode sentences. This patient had muscular atrophy, and they asked the patient to read the text on the screen aloud. They achieved a 25% word error rate with a 50-word vocabulary.

Here's a small tip: when judging whether a BCI paper is reliable, see whether the authors dare to release complete supplementary videos. That one trick filters out most of the fluff.

In 2023, UCSF added a digital avatar on top of this and expanded the vocabulary from 50 to 1,024, with almost no change in word error rate.

In the same year, on another electrode material, the Utah Array, the BrainGate team expanded the vocabulary from 50 to 125,000. Keep in mind that everyday English uses only about 10,000 common words, so 125,000 already includes many names and places. Their word error rate was 23.8%.

In 2024, the UCSF team built a bilingual BCI with a vocabulary of 104 and a word error rate of 25%.

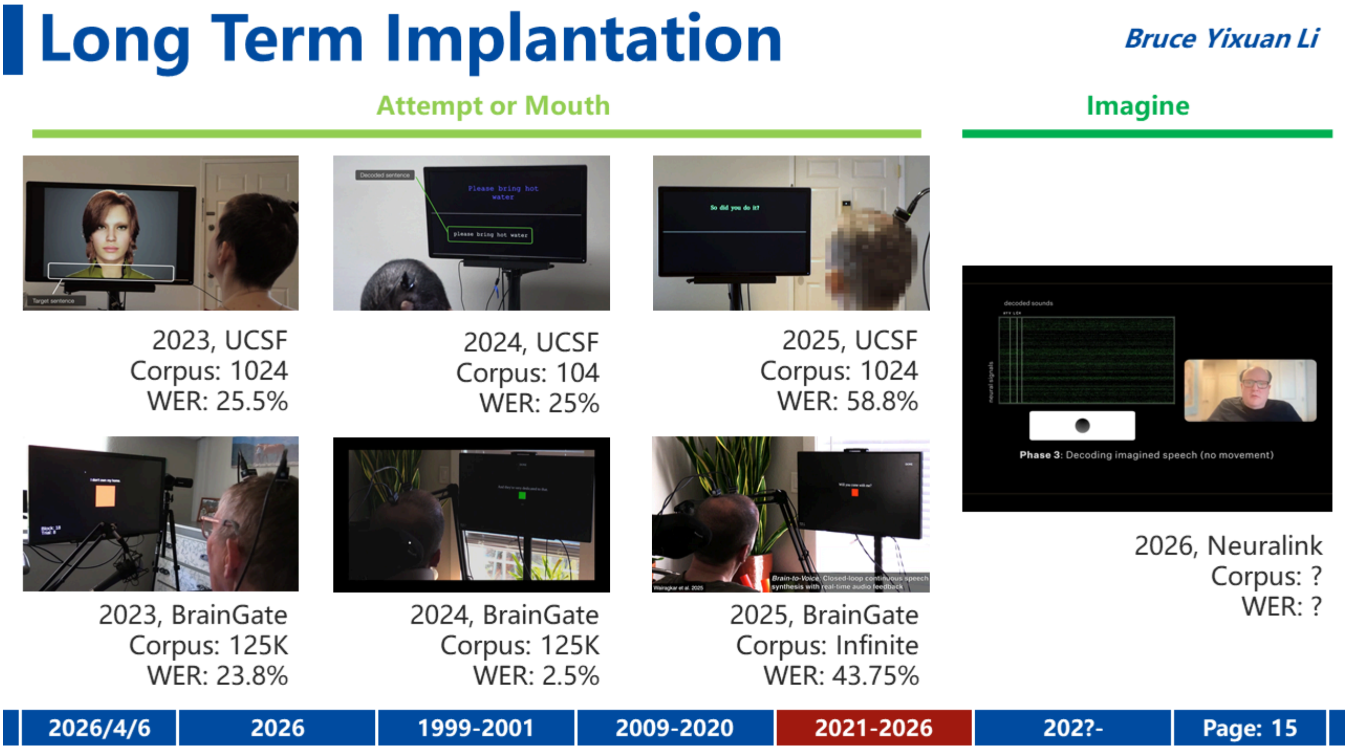
In the same year, the BrainGate team made an even bigger breakthrough: on a 125,000-word vocabulary, they reduced the word error rate to 2.5%. A key reason was that their participant was in the early stage of ALS and in better condition.

The 2021 and 2023 papers both used N-gram language models, while the two 2024 papers used deep-learning-based language models.

In 2025, the UCSF team achieved brain-to-voice, allowing the patient to control pitch, loudness, and speaking rhythm, but the word error rate rose to 50%.

In the same year, BrainGate also achieved brain-to-voice.

Listening to all this, you may feel that it's always just these two teams. And yes, before Neuralink joined, the only groups doing long-term implanted speech BCIs well were UCSF and BrainGate.



In 2026, Neuralink released a video, but they did not publish their vocabulary size or word error rate. Still, one major advance is clear: in the earlier papers, participants either read aloud or silently mouthed the words, whereas Neuralink reached pure imagined speech.

Three Stages

Bruce Yixuan Li

1999-2001

Which Signal? --- High Gamma

2009-2020

Which Model? --- LSTM

2021-2026

Safety?--- 5~10 Patients

There are now 5 to 10 patients with long-term implants, and their very existence already demonstrates the long-term safety of speech BCIs. The reason is simple: if the devices were not safe, these patients would not have made it this far.

The 4th Stage

Bruce Yixuan Li

1999-2001

Which Signal? --- High Gamma

2009-2020

Which Model? --- LSTM

2021-2026

How to Prove Safety? --- 5-10 Patients

202?-

Widely Adopted in Hospitals?

But at present, speech BCIs still have not entered major hospitals the way coronary bypass surgery has.

Widely Adopted in Hospitals?

Bruce Yixuan Li

1. What If The BCI Performance Go Down After 2 Years?

2. How to Solve High Variability Among Patients?

We Need A Fallback Algorithm.

In my view, there are two key problems that must be solved to get there in the future. First, after many years of implantation, when BCI performance declines, how can patients continue to benefit? Second, patients differ in disease severity, so the help provided by a BCI also differs. How can we ensure that implantation makes life easier for every patient, not just some of them?

In essence, both problems come down to designing a fallback algorithm. The ceiling is already very high: earlier work has enabled communication close to normal conversation. But the floor of speech BCIs has not yet been built.

Future

Bruce Yixuan Li

We Need A Fallback Algorithm.

1. Open Corpus -> Closed Corpus

2. Brain2Voice -> Brain2Text + TTS

3. Imagine -> Attempt

I see three ways to do this. First, give patients 1,000 English words or 1,000 Chinese characters, enough to handle communication about eating, drinking, toileting, and other daily needs. Second, there is no need to pursue flashy brain-to-audio systems; brain-to-text is enough, and TTS can handle the rest afterward. Third, we also do not need to rely on imagined speech. It is enough if the patient simply tries hard to speak.

Q & A

Bruce Yixuan Li

1. Would Speech BCI Reveal Thoughts that the Patients Does Note Want to Express?

2. Are There Any Public Datasets Available for Others to Validate?

New Videos on This Channel Will Address These Two Questions, as Well as Yours.

Any Questions Are Welcomed. Please Leave Them in the Comments.

After watching this, you may have two questions. Could this method expose thoughts the patient does not want to say out loud? And are there any public datasets that people can use for validation? A new video will address these two questions, and you are also welcome to leave your own questions in the comments.

Reference

Bruce Yixuan Li

1. Crone N. Functional mapping of human sensorimotor cortex with electrocorticographic spectral analysis. II. Event-related synchronization in the gamma band. *Brain*. 1998;121(12):2301-2315. doi:10.1093/brain/121.12.2301 **(The correlation of high-gamma signal and language)**
2. Crone NE, Boatman D, Gordon B, Hao L. Induced electrocorticographic gamma activity during auditory perception. *Clinical Neurophysiology*. 2001;112(4):565-582. doi:10.1016/S1388-2457(00)00545-9 **(The correlation of high-gamma signal and language)**
3. Makin JG, Moses DA, Chang EF. Machine translation of cortical activity to text with an encoder–decoder framework. *Nat Neurosci*. 2020;23(4):575-582. doi:10.1038/s41593-020-0608-8 **(The pinnacle of short-term implantation; vocabulary size of 250; word error rate of 3%; ECoG electrodes.)**
4. Moses DA, Metzger SL, Liu JR, et al. Neuroprosthesis for Decoding Speech in a Paralyzed Person with Anarthria. *N Engl J Med*. 2021;385(3):217-227. doi:10.1056/NEJMoa2027540 **(The first patient with long-term implantation who was able to decode hundreds of sentences.)**
5. Metzger SL, Littlejohn KT, Silva AB, et al. A high-performance neuroprosthesis for speech decoding and avatar control. *Nature*. 2023;620(7976):1037-1046. doi:10.1038/s41586-023-06443-4
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8. Silva AB, Littlejohn KT, Liu JR, Moses DA, Chang EF. The speech neuroprosthesis. *Nat Rev Neurosci*. 2024;25(7):473-492. doi:10.1038/s41583-024-00819-9 **(An excellent review article)**
9. Silva AB, Liu JR, Metzger SL, et al. A bilingual speech neuroprosthesis driven by cortical articulatory representations shared between languages. *Nat Biomed Eng*. 2024;8(8):977-991. doi:10.1038/s41551-024-01207-5
10. Littlejohn KT, Cho CJ, Liu JR, et al. A streaming brain-to-voice neuroprosthesis to restore naturalistic communication. *Nat Neurosci*. 2025;28(4):902-912. doi:10.1038/s41593-025-01905-6
11. Wairagkar M, Card NS, Singer-Clark T, et al. An instantaneous voice synthesis neuroprosthesis. *Cold Spring Harbor Laboratory*. Preprint posted online June 12, 2025. doi:10.1101/2024.08.14.607690

This is all for today.

Written Version of this Video

Bruce Yixuan Li

Chinese

https://github.com/Physics-Lee/20260315_Speech_Neuroprosthesis/blob/master/20260315_History_of_Speech_BCI/CN.pdf

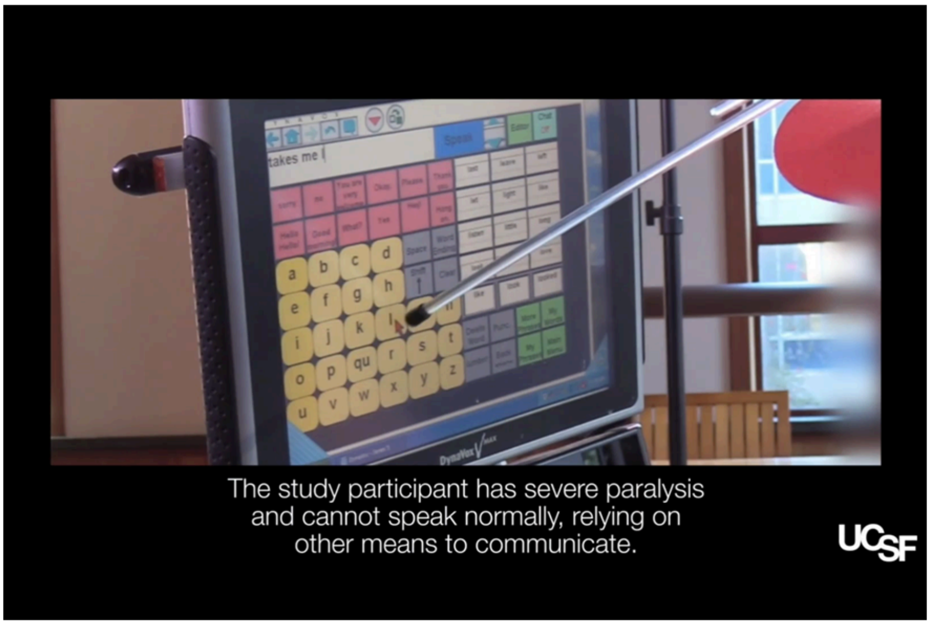
English

https://github.com/Physics-Lee/20260315_Speech_Neuroprosthesis/blob/master/20260315_History_of_Speech_BCI/EN.pdf

You can find the transcripts of the video here.

2021, UCSF

Bruce Yixuan Li



That concludes today's talk. The final minute is a 2021 UCSF promotional video. This was the world's first successful long-term implanted speech BCI to decode sentences.